

Digital Technologies in Education: Impacts and Factors influencing Digital Capacity and Transformation

Zair Bouzidi^{a,*}

^a*LMA Laboratory, University of Bejaia, Bejaia 06000, Algeria*

Abstract

Digital technologies have changed the nature and scope of education and led several education systems worldwide to adopt strategies and policies for ICT integration. The latter brought about issues regarding the quality of teaching and learning with ICTs, especially concerning the understanding, adaptation, and design of the education systems by current technological trends. These issues were emphasized during the recent COVID-19 pandemic that accelerated the use of digital technologies in education, generating questions regarding digitalization in schools. Specifically, many schools demonstrated a lack of experience and digital capacity, which resulted in widening gaps, inequalities, and learning losses. Such results have engendered the need for schools to learn and build upon the experience to enhance their digital capacity and preparedness, increase their digitalization levels, and achieve a successful digital transformation. Given that the integration of digital technologies is a complex and continuous process that impacts different actors within the school ecosystem, there is a need to show how these impacts are interconnected and identify the factors that can encourage effective and efficient change in the school environments. For this purpose, we conducted a non-systematic literature review. The results of the literature review were organized thematically based on the evidence they presented about the impact of digital technology on education and the factors that affect the schools' digital capacity and digital transformation. The findings suggest that

*Corresponding author

Email address: zair.bouzidi@univ-bejaia.dz (Zair Bouzidi)

the impact of ICT integration in schools goes beyond students' performance in school subjects and touches on several school-related aspects, such as equality and social integration, as well as different stakeholders. Additionally, various factors affect the impact of digital technologies in education. These factors are interconnected and play a vital role in the transformation process. The study results shed light on how ICTs can positively contribute to the digital transformation of schools and which factors should be taken into consideration for schools to achieve effective and efficient change.

Digital technologies have changed the nature and scope of education and have led several education systems around the world to adopt ICT integration strategies and policies. The latter has raised questions regarding the quality of teaching and learning with ICT, particularly regarding the understanding, adaptation and design of education systems in line with current technological trends. These issues have been highlighted during the recent Covid-19 pandemic which has accelerated the use of digital technologies in education, generating questions about digitalization in schools. Specifically, many schools have demonstrated a lack of experience and digital capacity, leading to widening gaps, inequalities and learning loss. Such findings have engendered the need for schools to learn and build on experience to improve their digital capacity and readiness, increase their levels of digitalization and achieve digital transformation. Given that the integration of digital technologies is a complex and continuous process that impacts different actors within the school ecosystem, it is necessary to show how these impacts are interconnected and to identify the factors that can promote effective and efficient change. effective in school environments. To do this, we carried out a non-systematic literature review. The results of the literature review were organized thematically based on the evidence they presented on the impact of digital technology on education and the factors that affect the digital capacity and digital transformation of schools. The results suggest that the impact of ICT integration in schools goes beyond students' performance in academic subjects and touches on several school-related aspects, such as equality and social integration, as well as as different stakeholders. Additionally, various

factors affect the impact of digital technologies in education. These factors are interconnected and play a vital role in the transformation process. The results of the study shed light on how ICT can positively contribute to the digital transformation of schools and what factors need to be considered for schools to achieve effective and efficient change.

Keywords: digital technologies, education, impact, digital capacity, digital transformation.

1 **Abbreviations list**

- 2 - API: Interface of Application Programming
- 3 - AI: Artificial Intelligence
- 4 - AR: Augmented Reality
- 5 - BERT: Representations of Bidirectional Encoder from Transformers
- 6 - CAI: Computer-Assisted Instruction
- 7 - CNN: Convolutional Neural Network
- 8 - FN: False Negative
- 9 - FP: False Positive
- 10 - GIS: Geospatial Information Technology
- 11 - GloVe: Global Vectors
- 12 - ICT: Information and Communication Technologies
- 13 - IoT: Internat of Things
- 14 - LDA: Latent Allocation of Dirichlet
- 15 - LSA: Latent Semantic Analysis
- 16 - LSTM: Long Short-Term Memory

- 17 - MCC: Matthews Correlation Coefficient
- 18 - MSA: Multi-Head Self-Attention
- 19 - NLP: Processing of Natural Language
- 20 - PCT: Position-wise Convolution Transformation
- 21 - RNN: Recurrent Neural Network
- 22 - SVM: Support Vector Machine
- 23 - TN: True Negative
- 24 - TP: True Positive
- 25 - US: United States
- 26 - VR: Virtual Reality

27 **Highlights**

- 28 - The study relies on a CNN-LSTM-based multimodal deep learning for
29 health surveillance
- 30 - It is based on multimodal deep learning, Bert, educational awareness, safe
31 news and re-emerging epidemics preventing
- 32 - It relies on CNN-LSTM-based multimodal deep learning, Bert, education,
33 and re-emerging epidemic preventing
- 34 - It examines CNN-LSTM-based multimodal deep learning, extended Bert,
35 education, and preventing fake news, in sustainability
- 36 - It It uses advancements in deep learning, Bert, connected objects, smart
37 data, and education to capture safe content



Figure 1: Graphical abstract

Graphical abstract

1. Introduction

Digital technologies have changed the nature and scope of education. Recently, versatile and disruptive technological innovations, such as smart devices, Internet of Things (IoT), artificial intelligence (AI), augmented reality (AR) and virtual reality (VR), blockchain, and software applications, have opened up new opportunities to advance teaching and learning (OECD, 2021; Gaol & Prasolova-Førland, 2021). Thus, in recent years, many education systems around the world have increased their investments in the integration of information and communication technologies (ICT) into teaching and learning practices [26, 45] (Fernández, Gimenez, & Calero, 2020; Lawrence & Tar, 2018) educational agendas the adaptation of an ICT integration strategy or policy [23] (European Commission, 2019). The latter has raised questions regarding the quality of teaching and learning with ICT [6] (Bates, 2015), particularly regarding the understanding, adaptation and design of education systems in line with current technological trends [5] (Balyer & Öz, 2018).

Specifically, previous studies have shown that despite the investments that several countries have made in the integration of technology in schools, the re-

sults are not promising and the expected results have not been achieved [18, 45] (Delgado et al., 2015; Lawrence & Tar, 2018). These issues have been highlighted during the recent COVID-19 pandemic as governments have been forced to move to online teaching at all levels of education [16] (Daniel, 2020). Online education has accelerated the use of digital technologies, generating questions about digitalization in schools [44] (König, Jäger-Biela, & Glutsch, 2020). Specifically, many schools have demonstrated a lack of experience and digital capacity, leading to widening gaps, inequalities, and learning losses [22] (European Commission, 2020). Such results have generated the need for schools to learn and build on the experience in order to improve their digital capacity and readiness [23] (European Commission, 2020) and increase their levels of digitalization [14] (Costa, Castaño Muñoz and Kampylis, 2021). Digitalization offers opportunities for fundamental improvements in schools [55] (Rott & Marouane, 2018; OECD, 2021) and affects many aspects of a school's development [17] (Decker & Ifenthaler, 2020). However, this is a complex process that requires large-scale transformative changes that go beyond the technical aspects of technology and infrastructure [51] (Pettersson, 2021).

Digitization can support a school's digital transformation, which refers to "a series of profound and coordinated changes in culture, workforce and technology and operating models" [8] (Brooks, & McCormack, 2020, p. 3) that bring cultural, organizational and operational change through the integration of digital technologies [42] (JISC, 2020).

For schools to succeed in their digital transformation, they must increase their digital capability levels. Digital capacity refers to "the extent to which culture, policies, infrastructure as well as the digital skills of students and staff support the effective integration of technology into teaching and learning practices" [14] (Costa, Castaño-Muñoz and Kampylis, 2021, p.163). Given that the integration of digital technologies is a complex and continuous process that impacts different actors within the school ecosystem [21] (Eng, 2005), it is necessary to show how these impacts are interconnected and to identify the factors that can encourage effective and efficient change in school environments. Based

87 on the above, this study aims to shed light on all the impacts of digital tech-
88 nologies on education and identify the factors that affect the digital capacity
89 and digital transformation of schools. Therefore, it will help to understand how
90 ICT can positively contribute to the digital transformation of schools. More
91 specifically, the document aims to answer the following questions:

- 92 a) What is the impact of digital technologies on education?
- 93 b) What factors could affect the digital capacity and transformation of
94 schools?

95 To answer the research questions, we conducted a non-systematic literature
96 review. The results of the literature review were organized thematically based on
97 the evidence they presented on the impact of digital technology on education
98 and the factors that affect the digital capacity and digital transformation of
99 schools.

100 **2. Methodology**

101 The non-systematic literature review presented here covers the main theories
102 and research published over the last 15 years on the subject. It was based
103 on meta-analyses and review articles found in peer-reviewed scientific content
104 databases and other key studies and reports related to the concepts studied (e.g.
105 digitalization, digital capability) professional and international organizations
106 (for example, the OECD).

107 Namely, we used the Scopus database to collect peer-reviewed academic arti-
108 cles. Scopus indexes various online education sector journals with international
109 reach. Additionally, we used the comprehensive Google Scholar search engine
110 to expand our searches based on relevant key terms or to include studies found
111 in the reference list of peer-reviewed articles, and other key studies and reports
112 linked to the concepts studied (e.g. digitalization, digital capacity) with pro-
113 fessional and international bodies. Finally, we gathered sources from the Pub-
114 lications Office of the European Union (<https://op.europa.eu/en/home>) to find
115 documents published by the EU that refer to policies related to digital transfor-

116 mation in education. First, we searched for resources on the impact of digital
 117 technologies in education by performing the following search queries: "impact"
 118 OR "effects" AND "digital technologies" AND "education", "impact" OR "ef-
 119 fects" AND "ICT" AND "education". We further refined our results by adding
 120 the terms "meta-analysis" and "review" or adjusting the search options based
 121 on the characteristics of each database to avoid collecting individual studies re-
 122 garding the use of one or more digital technologies in teaching and learning. We
 123 turned to meta-analyses and review studies because they consider the results of
 124 multiple studies providing important information about the state of research in
 125 a given area [57] (Schuele & Justice, 2006). Individual studies were excluded
 126 because they make only limited contributions to a particular field. Additionally,
 127 effect size information reported in meta-analyses provides important insight into
 128 the extent of the possible impact of digital technology use on learning [37] (Hig-
 129 gins et al., 2012). Although quantitative data is important, it does not provide
 130 explanations for the challenges or difficulties encountered when integrating ICT
 131 into learning and teaching [64] (Tolani-Brown, McCormac, & Zimmermann,
 132 2011). In this regard, literature reviews were valuable sources as they provided
 133 a detailed understanding of the benefits and implications of integrating tech-
 134 nology in schools. Primarily, the review studies provided in-depth qualitative
 135 evidence between the use of digital technology or self-reflection tools and their
 136 outcomes. In contrast, effect factors derived from meta-analyses of experimental
 137 studies have provided quantitative evidence since they involve a measurement of
 138 impact based on statistically verifiable results [64] (Tolani Brown, McCormac,
 139 & Zimmermann, 2011).

140 In the analysis presented here, we have also included policy documents and
 141 reports from professional and international bodies and government reports to
 142 gain a valuable understanding of this study's key concepts and policy recommen-
 143 dations regarding technology integration digital technology in education. These
 144 papers and reports have provided useful insights into recent and important evi-
 145 dence regarding the concept of digital capacity and educational transformation.
 146 The inclusion and exclusion criteria are presented in Table 1.

147 Table 1 Inclusion and exclusion criteria regarding the impact of digital tech-
148 nologies in education.

149 Inclusion criteria
150 papers since 2005
151 review and meta-analysis studies
152 formal education K-12
153 peer-reviewed articles
154 articles in English
155 professional/international bodies, and government reports
156 book chapters
157 Exclusion criteria
158 Theses and doctoral theses
159 posters in conferences
160 conference papers without proceedings
161 sources on higher education

162 To ensure reliable extraction of information from each study and to assist
163 in research synthesis, we selected study characteristics of interest (impacts) and
164 constructed coding forms. First, an overview of the synthesis was provided by
165 the principal investigator who described the processes of coding, data entry, and
166 data management. All coders worked independently based on common instruc-
167 tions. To ensure a common understanding of the process among coders, the
168 sample of ten studies was tested. The results were compared and discrepancies
169 were identified and resolved. Additionally, to ensure an efficient coding process,
170 all coders participated in group meetings to discuss additions, deletions, and
171 modifications [59] (Stock, 1994).

172 Due to the methodological diversity of the documents studied, we began to
173 synthesize the results of the literature review based on similar study designs.
174 Specifically, most meta-analysis studies were grouped into a single category
175 due to the fact that impact was measured quantitatively in the form of effect
176 sizes that primarily related to student achievement [35] (Hattie, Rogers and
177 Swaminathan, 2014). Next, we organized the themes of qualitative studies on

178 other categories based on their impacts. Finally, we synthesized data from
179 reviews and meta-analyses across categories.

180 To establish a collective understanding of the concept of impact, we referred
181 to a previous impact study which was conducted [4] by Balanskat (2009) with
182 a similar scope. The aforementioned study examined the impact of technology
183 in primary schools. In this context, impact had a more specific meaning related
184 to ICT and was described as “a significant influence or effect of ICT on the
185 measured or perceived quality of (parts of) education” [4] (Balanskat, 2009, p.
186 9). In the study presented here, the main impacts refer to learning and learners,
187 teaching and teachers, as well as other key stakeholders who are directly or
188 indirectly linked to the academic unit.

189 3. Conclusion

190 The results of the study showed the impacts of the use of digital technolo-
191 gies on the knowledge, skills and attitudes of students, on equality, inclusion
192 and social integration, on the professional and educational practices of stu-
193 dents, teachers and on other aspects and actors linked to school. The data also
194 reported on various factors that could affect the digital capacity and transforma-
195 tion of schools, such as digital skills, personal characteristics and professional
196 development of teachers, school leadership and management, administration,
197 infrastructure, etc. The impacts and factors found in the literature review are
198 presented below.

199 3.1. *Impacts of digital technologies on students’ knowledge, skills and attitudes*

200 The impact of ICT use on students’ knowledge, skills and attitudes has been
201 studied very early in the literature. Eng (2005) [21] found a slight positive effect
202 between ICT use and student learning.

203 Specifically, the author reported that access to computers for each student
204 could improve learning when used in computer-assisted instruction (CAI) pro-
205 grams in simulation or tutoring modes that are used to complement rather

206 than replace teaching. The author reported studies that showed that teach-
 207 ers recognized the benefits of ICT for students with special educational needs;
 208 however, the impact of ICT on student achievement was unclear. Balanskat,
 209 Blamire & Kefala (2006) found a statistically significant positive association
 210 between ICT use and higher student achievement in primary and secondary
 211 education. The authors also reported improvements in the performance of
 212 low-performing students. The use of ICT resulted in other positive gains for
 213 students, namely increased attention, engagement, motivation, communication
 214 and process skills, teamwork and increased learning behavior. Evidence re-
 215 ported from qualitative studies showed that teachers, students and parents rec-
 216 ognized the positive impact of ICT on student learning, regardless of their pro-
 217 ficiency level (strong/weak students). Punie, Zinnbauer, and Cabrera (2006)
 218 documented studies that showed positive results of ICT-based learning to sup-
 219 port low-achieving students and young people with complex lives outside of the
 220 education system. Liao, Chang, and Chen (2007) reported moderate positive
 221 effects of teaching computer applications (computer-assisted instruction, com-
 222 puter simulations, and Web-based learning) compared to traditional teaching
 223 on elementary school students' achievement. Similarly, Tamim et al. (2011)
 224 reported small to moderate positive effects between the use of computer tech-
 225 nology (CAE, ICT, simulations, computer-assisted instruction, digital and hy-
 226 permedia) and student achievement in formal face-to-face classes compared to
 227 classes that did not use technology. . Jewitt and Hadjithoma Garstka (2011)
 228 found that the use of learning platforms (LPs) (virtual learning environments,
 229 management information systems, communication technologies, and informa-
 230 tion and resource sharing technologies) in schools provided more opportuni-
 231 ties for primary and secondary students to access a variety and quality learn-
 232 ing resources, engage in independent and personalized learning, conduct self-
 233 assessment and peer review, as well as opportunities for teacher evaluation and
 234 feedback. Similar findings were reported by Fu (2013), who documented a list
 235 of benefits and opportunities of using ICT.

236 According to the author, the use of ICT helps students effectively and effi-

237 ciently access digital information and course content, supports student-centered
238 and self-directed learning, supports the development of an environment creative
239 learning where more opportunities for critical thinking skills are offered and
240 promotes collaborative learning in a distance learning environment. Higgins et
241 al. (2012) found consistent but weak positive associations between technology
242 use and learning outcomes of school-aged learners (5-18 years) in studies linking
243 supply and use technology and level of success. Additionally, Chauhan (2017)
244 reported a medium positive effect of technology on the learning effectiveness of
245 primary school students compared to students who received traditional educa-
246 tion.

247 The rise of mobile technologies and hardware devices has led to new research
248 into their impact on teaching and learning. Sung, Chang, and Liu (2016) re-
249 ported a moderate effect on student performance of using mobile devices in the
250 classroom compared to using desktop computers or not using mobile devices.
251 Schmid et al. (2014) reported medium to small positive effects of technology
252 integration (e.g., CAI, ICT) in the classroom on student achievement and at-
253 titudes compared to no technology use or its use to varying degrees. Tamimi
254 et al. (2015) found a small, statistically significant effect of using tablets and
255 other smart devices in educational settings on students' academic achievement.
256 The authors also reported that tablets provided additional benefits to students;
257 namely, they reported improvements in students' note-taking, organizational
258 and communication skills, and creativity. Zheng et al. (2016) reported a small
259 positive effect of individual laptop programs on students' academic achievement
260 in all subjects. Other reported benefits included student-centered, individual-
261 ized, and project-based learning, which increased learner engagement and enthu-
262 siasm. Additionally, the authors found that students in individual laptop-based
263 programs tended to use technology more frequently than in classes without lap-
264 tops and, as a result, they developed a range of skills (e.g., information, media
265 , technological skills, organization skills). Haßler, Major, and Hennessy (2016)
266 found that most interventions that included the use of tablets across the curricu-
267 lum reported positive learning outcomes. However, out of twenty-three studies,

268 five studies reported no difference and two reported a negative effect on student
269 learning outcomes with the use of tablets.

270 Recently, approaches referring to the impact of gamification with the use of
271 digital technologies on teaching and learning have also been explored.

272 Huang et al. (2020) reported a moderate effect of gamification (features of
273 video games) on student learning outcomes in formal education contexts com-
274 pared to non-gaming contexts. Talan, Doğan, and Batdı (2020) reported a
275 medium effect of using educational games on academic achievement. Over the
276 past two years, the effects of more advanced technologies on teaching and learn-
277 ing have also been studied. Garzón & Acevedo (2019) found that augmented
278 reality (AR applications) had a medium effect on student learning outcomes
279 compared to traditional lectures. Similarly, Garzón et al. (2020) showed that
280 AR had a medium impact on student learning gains.

281 It should be noted that the vast majority of studies refer to learning gains
282 in specific subjects. Specifically, several studies have examined the impact of
283 digital technologies on students' literacy skills and reported positive effects on
284 language learning (Balanskat et al., 2006; Grgurović, Chapelle, & Shelley, 2013;
285 Friedel et al. al., 2013; Furthermore, several studies have documented positive
286 studies on specific areas of language learning, namely foreign language learning
287 (Kao, 2014), writing (Higgins et al., 2012; Zheng et al., 2016), reading and com-
288 prehension (Liao et al., 2007; Cheung and Slavin, 2012). ICT also has a positive
289 impact on students' performance in mathematics (Balanskat et al., 2006; Li &
290 Ma, 2010; Hillmayr et al., 2010; Zheng et al., 2016; Verschaffel, Depaepe, &
291 Mevarech, 2019). More specific positive learning gains have been reported in
292 numerical skills, problem-solving skills and pattern exploration skills (Higgins et
293 al., 2012) and metacognitive learning outcomes (Verschaffel et al. , 2019). In ad-
294 dition to learning gains, studies have also reported improvements in motivation
295 and interest (Higgins et. al., 2012) as well as increased confidence and engage-
296 ment in mathematics (Huang et. al., 2012). al., 2014). Furthermore, study
297 results have documented the positive effects of ICT on science learning (Balan-
298 skat et al., 2006; Liao et al., 2007; Zheng et al., 2016; Hillmayr et al., 2020).

299 Finally, some studies have documented that the use of ICT leads to positive
300 impacts on student achievement in other subjects, such as geography, history
301 and the arts (Condie and Munro, 2007), design and technology (Balanskat et
302 al., 2006) and music and arts (Chauhan, 2017).

303 3.2. Impacts of digital technologies on equality, inclusion and social inte-
304 gration Although most of the studies reviewed have focused on the impact of
305 ICT on students' knowledge, skills and attitudes, some reports have also been
306 written on other aspects that may be impacted in the school context, such as
307 equality, inclusion and social integration. Condie and Munro (2007) documented
308 research interventions examining how ICT can support students with additional
309 or special educational needs. Although these interventions were conducted on
310 a relatively small scale and primarily based on qualitative data, their findings
311 indicated that ICT use improved communication, participation and self-esteem.
312 Underwood (2009) reported that the use of digital technologies enhanced social
313 interaction and improved academic performance of children with learning diffi-
314 culties. Istenic and Bagon (2014) found that the role of ICT in inclusion and
315 the design of educational and technological interventions was not sufficiently
316 explored in educational interventions; however, some benefits of ICT have been
317 found in the social integration of students. The issue of gender equality was
318 mentioned in a small number of studies. Zheng et al. (2016) reported a statis-
319 tically significant positive interaction between individual laptop programs and
320 gender.

321 Specifically, the results showed that both girls and boys benefited from the
322 laptop program, but the effect on girls' achievement was smaller compared to the
323 effect on boys' achievement. Results of a systematic review by Cussó-Calabuig,
324 Farran, and Bosch-Capblanch (2018) found no evidence that heavy computer
325 use can reduce gender differences in anxiety, self-efficacy and self-confidence.

326 *3.2. Impacts of digital technologies on the professional and educational practices*
327 *of teachers*

328 There is evidence from various research studies on the impact of ICT on
329 teachers' teaching practices and student assessment. Fridel et al. (2013) found
330 that student use of mobile devices enabled teachers to successfully deliver con-
331 tent (e.g., mobile serious games), lessons, and scaffolding and facilitated syn-
332 chronous collaborative learning. Furthermore, ICT can increase the effectiveness
333 of lesson planning and preparation by providing opportunities for a more col-
334 laborative approach between teachers. Sharing curricula and analyzing student
335 data resulted in clearer goals and improved reporting to parents (Balanskat et
336 al., 2006). Additionally, the use and application of digital technologies in teach-
337 ing and learning can improve teachers' digital skills. Balanskat et al. (2006)
338 documented various studies which found that the use of digital technologies
339 in education had positive effects on teachers' basic ICT skills. The greatest
340 impact was seen among teachers with sufficient experience in integrating ICT
341 into their teaching and/or having recently participated in advanced courses in
342 the pedagogical use of technologies in teaching. Punie, Zinnbauer, and Cabrera
343 (2006) reported that the provision of fully equipped multimedia laptops and
344 the development of online teacher communities had positive effects on teachers'
345 confidence and competence in using ICT.

346 Additionally, the use of ICT for online assessment benefits teaching. In
347 particular, online assessments support the digitalization of student work and
348 associated logistics, enable teachers to gather immediate feedback and readjust
349 to new goals, and support the improvement of the technical quality of tests
350 providing more accurate results. In addition, ICT capabilities (p. productive in
351 groups (Punie, Zinnbauer, & Cabrera, 2006).

352 *3.4 Impacts of digital technologies on other aspects and actors linked to*
353 *school* There is evidence that the effective use of ICT and the data transmission
354 offered by broadband connections have contributed to improved administration
355 (Balanskat et al., 2006). Specifically, ICT provides better management systems
356 to schools that enable data collection procedures. Condie and Munro (2007)

357 listed the improvements made to the school through the use of ICT in the
 358 following areas: attendance monitoring, assessment records, reporting to par-
 359 ents, financial management, creation of benchmarks for learning resources and
 360 information sharing among staff. This data can be used strategically for self-
 361 assessment and monitoring purposes which, in turn, can lead to improvements at
 362 the school. Furthermore, ICT has provided more efficient and effective examina-
 363 tion management procedures, namely less time-consuming reporting processes
 364 than paper-based examinations and smooth communications between schools
 365 and examination authorities through the exchange of electronic data (Punie,
 366 Zinnbauer and Cabrera, 2006). Zheng et al. (2016) reported that the use of
 367 ICT improved home-school relationships. Additionally, Escueta et al. (2017) re-
 368 ported several ICT programs that improved the flow of information from school
 369 to parents.

370 In particular, they documented that the use of ICT (learning management
 371 systems, emails, dedicated websites, mobile phones) provided opportunities for
 372 personalized and personalized information exchange between schools and par-
 373 ents , such as attendance records, upcoming assignments, school events, stu-
 374 dent grades, which generated positive results on learning outcomes and student
 375 achievement. Specifically, the exchange of information between schools and
 376 families motivated parents to encourage their children to exert effort in school.

377 The above findings suggest that the impact of ICT integration in schools
 378 goes beyond students' performance in academic subjects. Concretely, it affects
 379 a certain number of aspects linked to school, such as equality and social integra-
 380 tion, professional and educational practices as well as the various stakeholders.
 381 In Table 2, we summarize the different impacts of digital technologies on school
 382 stakeholders based on the literature review. Table 2 The impact of digital tech-
 383 nologies on school stakeholders based on the literature review Actors of the
 384 Impacts school Students Improvements in knowledge, skills and attitudes

385 Positive learning gains from the use of ICT across the curriculum

386 Positive learning gains from the use of ICT in specific school subjects (e.g.
 387 mathematics, literacy, science)

388 Positive learning gains for students with special needs and low-achieving
389 students

390 Developing a range of skills (e.g. developing subject-related, communication,
391 negotiation, organizational, critical thinking, creativity skills)

392 Development of digital skills (e.g. information skills, media skills, ICT skills)

393 Positive attitudes and behaviors towards ICT (e.g. increased interest, moti-
394 vation, attention, engagement, confidence)

395 Improved learning experience

396 Better access to resources

397 Opportunities to experiment with various learning practices (e.g. active
398 learning, learner-centered learning, independent and personalized learning, col-
399 laborative learning, self-directed learning, self-assessment and peer assessment)

400 Improving access to teacher assessment and feedback

401 Support for equality, inclusion and social integration

402 Improving communication, participation and self-esteem of students with
403 special needs.

404 Improved social interaction for students in general and for students with
405 learning difficulties.

406 Improving the success of girls and boys through the use of ICT.

407 Teachers

408 Improvement of professional practice

409 Development of digital skills

410 Positive attitudes and behaviors towards ICT (e.g. increased confidence).

411 Formalized collaborative planning between teachers.

412 Improved relationships with parents.

413 Improvement of teaching practice

414 Efficiency in lesson planning and preparation

415 Effective Ways to Provide Feedback

416 Improvements resulting from readjustment of educational objectives

417 Improvements to the technical quality of testing

418 Practice new methods for testing specific skills (e.g., problem-solving skills,
 419 metacognitive skills)

420 Content delivery and successful lessons

421 Application of different teaching practices (e.g. scaffolding, synchronous
 422 collaborative learning)

423 Administrators

424 Improved data-driven decision making

425 Improved data collection processes

426 Improved monitoring and evaluation processes (e.g. attendance tracking,
 427 financial management, evaluation records)

428 Improvement of organizational processes

429 Improving access to learning resources through the creation of repositories

430 Improved information sharing between school staff

431 Smooth communications with external authorities (e.g. review results)

432 Effective and successful exam management procedures

433 Improved communication between home and school

434 Improved relationships with parents.

435 Improved communication flow between school and parents (e.g. personalized
 436 and personalized communications).

437 Parents

438 Improved home-school relationships

439 Increased parental participation in children's school life

440 Heads of establishment

441 Improvement of professional practice

442 Reduction of the isolation of the head of establishment.

443 Improving access to information about school improvement practices.

444 Improved exchange of information on diverse perspectives on key issues.

445 *3.3. Factors that influence the integration of digital technologies*

446 Several authors have suggested that the impact of technologies in education
 447 depends on several factors and not on the technology itself. For example, Liao

et al. (2007) suggested that future studies should carefully examine the factors that contribute to positive outcomes in clarifying the exact relationship between computer applications and learning. Additionally, Haßler, Major, and Hennessy (2016) suggested that the neutral findings regarding the impact of tablets on student learning outcomes should encourage educators, school leaders, and school officials to further study the potential of these devices in teaching and learning. Several other researchers have suggested that various variables play an important role in the impact of ICT on student learning, which could be attributed to context, teaching and professional development practices, curriculum and learner characteristics (Underwood, 2009; Tamim, 2011; Higgins et al., 2012; Archer and Savage, 2014; Sung et al., 2016; Haßler, Major, and Hennessy, 2016; Chauhan, 2017; Lee et al., 2020).

3.3.1. *Digital skills*

One of the most common challenges reported in studies using digital tools in the classroom was students' lack of skills on how to use them.

Fu (2013) found that students' lack of technical skills is a barrier to the effective use of ICT in the classroom. Tamimi et al. (2015) reported that students experienced difficulties when using tablets and smart mobile devices, which were associated with the technical issues or expertise required to use them, the distracting nature of the devices, and highlighted the need for professional development of teachers.

Higgins et al. (2012) reported that skills training on the use of digital technologies is essential for learners to fully exploit the benefits of teaching.

On another note, Delgado et al. (2015) reported studies that showed a strong positive association between teachers' computer skills and students' computer use. With this in mind, teachers' lack of ICT skills and familiarity with technologies could become an obstacle to effective use of technology in the classroom (Balanskat et al., 2006; Delgado et al., 2015). It should be noted that the way teachers are introduced to ICT affects the impact of digital technologies on education. Previous studies have shown that teachers may avoid using digital

478 technologies due to limited digital skills (Balanskat, 2006), or that they prefer
479 to apply "safe" technologies, i.e. technologies that their own teachers have used
480 and with which they are familiar (Condie and Munro, 2007).

481 In this regard, providing digital skills training and exposure to new digital
482 tools could encourage teachers to apply various technologies in their lessons
483 (Condie and Munro, 2007). In addition to digital competence, school-based
484 technical support has also been shown to affect teachers' use of technology in
485 their classrooms (Delgado et al., 2015). Providing support can reduce time
486 and cognitive constraints, which could result in limited integration of ICT into
487 school lessons by teachers (Escueta et al., 2017).

488 3.3.2. *Personal characteristics, training approaches and professional develop-* 489 *ment of teachers*

490 Teachers' personal characteristics and professional development affect the
491 impact of digital technologies in education. Specifically, Cheok and Wong (2015)
492 found that teachers' personal characteristics (e.g., anxiety, self-efficacy) are as-
493 sociated with teachers' satisfaction and engagement with technology. Bingim-
494 las (2009) reported that lack of confidence, resistance to change and negative
495 attitudes towards the use of new technologies in teaching are important deter-
496 minants of the level of teachers' engagement with ICT. The author reports that
497 providing technical support, motivational support (e.g., rewards, sufficient time
498 for planning), and training on how technologies can benefit teaching and to
499 learning can remove the barriers to ICT integration mentioned above.

500 Archer and Savage (2014) found that comfort with technology is a signifi-
501 cant predictor of technology integration and argued that it is essential to provide
502 teachers with appropriate training and ongoing support until that they are com-
503 fortably with the use of ICT in class. Hillmayr et al. (2020) documented that
504 teacher training in ICT had a significant effect with the use of ICT on student
505 learning.

506 According to Balanskat et al. (2006), the impact of ICT on student learning
507 strongly depends on the teacher's ability to use it effectively for educational

508 purposes. The results obtained from the Teaching and Learning International
509 Survey (TALIS) (Organization for Economic Co-operation and Development,
510 2020) revealed that although schools are open to innovative practices and have
511 the capacity to adopt them, only 39% of teachers in the European Union say
512 they are well or very well prepared to use digital technologies for teaching.

513 Li and Ma (2010) and Hardman (2019) showed that the positive effect of
514 technology on student achievement depended on the instructional practices used
515 by teachers. Schmid et al. (2014) reported that learning was best supported
516 when students were engaged in active, meaningful activities via technology tools
517 that provided cognitive support.

518 Tamimi et al. (2015) compared two different instructional uses of tablets and
519 found a significant moderate effect size when the devices were used in a student-
520 centered context and approach rather than teacher-directed environments.

521 Similarly, Garzon et al. (2019) and Garzon et al. (2020) reported that the
522 positive results of integrating AR applications could be attributed to the exis-
523 tence of different variables that could influence AR interventions (e.g., teaching
524 approach, learning environment and the duration of the intervention).

525 Additionally, the authors suggested that the instructional resources that
526 teachers used to supplement their lessons and the instructional approaches they
527 applied were critical to the effective integration of AR in education. Hattie
528 (2009) reported that effective use of computers is associated with training teach-
529 ers to use computers as a teaching and learning tool. Zheng et al. (2016) noted
530 that in addition to the strategies teachers adopted in teaching, ongoing profes-
531 sional development was also essential to ensure the success of technology imple-
532 mentation programs. Song et al. (2016) found that most research on the use of
533 mobile devices to support learning indicated that insufficient teacher prepara-
534 tion was a major barrier to implementing effective mobile learning programs in
535 schools. Friedel et al. (2013) found that teacher training and support increased
536 the positive impact of interventions on student learning gains.

537 Trucano (2005) argued that positive impacts occur when digital technologies
538 are used to enhance a teacher's existing instructional philosophies. Higgins et

al. (2012) found that the types of technologies used and how they are used can also affect student learning. The authors suggest that teacher training and professional development focused on the effective pedagogical use of technology to support teaching and learning is an important component of successful teaching approaches (Higgins et al., 2012). Archer et al. (2014) found that studies reporting ICT interventions in which teachers received training and support had moderate positive effects on student learning outcomes, which were significantly higher than studies where little or no details of training and support were mentioned. Fu (2013) reported that teachers' lack of knowledge and skills on the technical and pedagogical aspects of using ICT in the classroom, in-service training, teaching support, technical and financial support, as well as the lack of motivation and encouragement to integrate ICT into their teaching constituted significant obstacles to the integration of ICT into education.

3.3.3. *School leadership and management*

Management and leadership are important cornerstones of the digital transformation process (Pihir, Tomičić-Pupek, & Furjan, 2018).

Zheng et al. (2016) documented leadership among the factors that positively impacted the successful implementation of technology integration in schools.

Strong leadership, strategic planning and systematic integration of digital technologies are prerequisites for the digital transformation of education systems (Redep, 2021). Management and leadership play an important role in formulating policies that translate into practice and ensure that ICT developments are integrated into school life and the experiences of staff and students (Condie & Munro, 2007). Policy support and leadership involves providing a comprehensive vision for the use of digital technologies in education, student and parent guidance, logistical support, and teacher training (Conrads, 2014). Unless there is whole-school engagement with accountability for progress at key points, it is unlikely that ICT integration will be sustained or become part of the culture (Condie and Munro, 2007). To this end, school leaders must adopt and promote a whole-school strategy and build a strong mutual support system

569 that enables the technological maturity of the school (European Commission,
570 2019). In this context, school culture plays an essential role in shaping the men-
571 talities and beliefs of all school stakeholders towards successful technological
572 integration.

573 Condie and Munro (2007) highlighted the importance of the principal's en-
574 thusiasm and hard work in inspiring school staff and students to cultivate a
575 culture of innovation and achieve lasting digital change.

576 Specifically, school leaders must create conditions in which school staff are
577 empowered to experiment and take risks with technology (Elkordy and Lovinelli,
578 2020).

579 For leaders to achieve the above, it is important to develop learning and lead-
580 ership capabilities, advocate professional learning and create support systems
581 and structures (European Commission, 2019). Integrating digital technology
582 into education systems could be difficult and leaders need guidance to achieve
583 this. Guidance can be introduced by adopting new strategic planning methods
584 and techniques for the integration of digital technologies (Redep, 2021). Al-
585 though the role of leaders is essential, their training is not as frequent as it
586 should be. More specifically, only a third of education systems in Europe have
587 implemented national strategies that explicitly refer to the training of school
588 leaders (European Commission, 2019, p. 16).

589 3.3.4. *Connectivity, infrastructure and government and other support*

590 The effective integration of digital technologies at all levels of education
591 requires the development of infrastructure, the provision of digital content and
592 the selection of suitable resources (Voogt et al., 2013).

593 In particular, a high-quality broadband connection at school increases the
594 quality and quantity of educational activities. There is evidence that ICT in-
595 creases and formalizes cooperative planning among teachers and cooperation
596 with managers, which in turn has a positive impact on teaching practices (Bal-
597 anskat et al., 2006). Additionally, ICT resources, including software and hard-
598 ware, increase the opportunity for teachers to integrate technology into the

599 curriculum to improve their teaching practices (Delgado et al., 2015). For ex-
600 ample, Zheng et al. (2016) found that the use of individual laptop programs led
601 to positive changes in teaching and learning, which would not be accomplished
602 if no infrastructure and technical support were provided to teachers. Delgado
603 et al. (2015) reported that limited access to technology (insufficient computers,
604 peripherals and software) and lack of technical support are significant barriers
605 to ICT integration. Access to infrastructure refers not only to the availabil-
606 ity of technology in a school, but includes providing the appropriate amount
607 and types of technology in places where teachers and students can use them.
608 Effective technical support is a central part of the school's overall ICT strat-
609 egy (Underwood, 2009). Bingimlas (2009) reported that the lack of technical
610 support in the classroom and school-wide resources (e.g. by teachers.

611 Additionally, poor and inadequate hardware maintenance and unsuitable
612 educational software can discourage teachers from using ICT in education (Bal-
613 anskat et al., 2006; Bingimlas, 2009).

614 Government support can also have an impact on the integration of ICT in
615 education. More specifically, Balanskat et al. (2006) reported that government
616 interventions and training programs increased teachers' enthusiasm and positive
617 attitudes towards ICT and led to the systematic use of integrated ICT.

618 Finally, another important factor affecting digital transformation is the de-
619 velopment and quality assurance of digital learning resources. These resources
620 may be supporting textbooks and related materials or resources that focus on
621 specific topics or parts of the curriculum. Policies relating to the provision
622 of digital educational resources are essential for schools and can be achieved
623 through various actions.

624 For example, some countries fund web portals that become repositories,
625 allowing teachers to share resources or create their own. Additionally, they may
626 offer online learning opportunities or other services related to digital education.
627 In other cases, specific project agencies have also been set up to develop digital
628 resources (Eurydice, 2019).

629 *3.3.5. Administration and management of digital data*

630 The digital transformation of schools requires organizational improvements
631 in internal workflows, communication between different stakeholders and collab-
632 oration possibilities. Vuorikari, Punie, and Cabrera (2020) presented evidence
633 that digital technologies supported the automation of administrative practices
634 in schools and reduced administrative overload. There is evidence that digital
635 data affects the production of knowledge about schools and has the power to
636 transform the way schooling takes place. Specifically, Sellar (2014) reported
637 that data infrastructure in education is growing due to the demand for "in-
638 formation on student achievement, teacher quality, academic performance, and
639 adult skills, associated with policy efforts to increase human capital and pro-
640 ductivity practices" (p. 771).). In this regard, practices such as datafication
641 which refers to the translation of information about all kinds of things and pro-
642 cesses into quantified formats have become essential for decision-making based
643 on quality accountability data reporting. of school. The use of data could be
644 transformed into in-depth knowledge about education or training through the
645 use of ICT. For example, measuring students' online engagement with learning
646 materials and drawing meaningful conclusions can allow teachers to improve
647 their instructional interventions (Vuorikari, Punie, & Cabrera, 2020).

648 *3.3.6. Socio-economic origin of students and family support*

649 Research shows that parents' active engagement in the school and their
650 support of the school's work can make a difference to their children's attitudes
651 towards learning and, therefore, their success (Hattie, 2008) . In recent years,
652 digital technologies have been used for more effective communication between
653 school and family (Escueta et al., 2017). The European Commission (2020)
654 presented data from a recent Eurostat survey regarding students' computer use
655 during the pandemic. The data showed that younger students needed additional
656 support and guidance from parents and that challenges were greater for families
657 whose parents had lower levels of education and digital skills. zero or weak.

658 In this regard, the socio-economic background of learners and their socio-

cultural environment also affect academic performance (Punie, Zinnbauer, & Cabrera, 2006).

Trucano documented that computer use at home positively influenced student confidence and resulted in more frequent use in schools compared to students who did not have access at home (Trucano, 2005). In this sense, socio-economic background affects access to computers at home (OECD, 2015) which in turn influences the experience of ICT, which is an important factor for academic success (Punie, Zinnbauer, & Cabrera, 2006;

Furthermore, parents from different socio-economic backgrounds may have different abilities and availability to support their children in their learning process (Di Pietro et al., 2020).

3.3.7. Socio-economic school context and emergency situations

The socio-economic context of the school is closely linked to the digital transformation of the school. For example, schools located in disadvantaged, rural or disadvantaged areas are likely to lack the appropriate digital capacity and infrastructure needed to adapt to the use of digital technologies during emergency periods, such as than the COVID-19 pandemic (Di Pietro et al., 2020). Data collected from school principals confirms that in several countries there is a rural/urban divide in connectivity (OECD, 2015).

Emergency periods have also affected the digitalization of schools. The Covid-19 pandemic has led to the closure of schools and forced them to look for various appropriate and connective ways to continue working on the curriculum (Di Pietro et al., 2020). The sudden large-scale shift to distance and online teaching and learning has also raised various challenges regarding the quality and equity of education, such as the risk of increasing learning inequalities, digital and social, and the difficulties encountered by teachers in dealing with them. demanding situation (European Commission, 2020). Looking at the results of the above studies, we can conclude that the impact of digital technologies in education refers to various stakeholders and touches many aspects of the school ecosystem. Figure 1 summarizes the factors that affect the impact of digital

689 technologies on school stakeholders based on the results of the literature review.

690 Fig.1 Factors affecting the impact of ICT on education

691 4. Discussion

692 The results revealed that the use of digital technologies in education affects
693 different actors within a school ecosystem. First, we observed that as technolo-
694 gies evolve, the interest of the research community in applying them in schools
695 also increases. Starting as early as 2005 where the use of interactive whiteboards
696 was most commonly applied to school interventions (references), evolving to-
697 wards the use of mobile devices and digital games (references) to more recent
698 technological advances, such as applications AR and VR (references), technol-
699 ogy has always been applied to improve school life. This indicates that the value
700 of technologies to support teaching and learning as well as other school-related
701 practices is recognized by the research and school community. The impact re-
702 sults of the literature review indicated that ICT integration has weak effects
703 (Eng, 2005; Zheng et al., 2016; Higgins et al., 2012; Schmid et al., 2014; Tamim
704 et al. al., 2015) and moderate (Liao et al., 2007 ; Sung et al., 2016 ; Huang et
705 al., 2020 ; Talan et al., 2020 ; Garzón & Acevedo, 2019 ; Garzón et al., 2020)
706 on student learning outcomes.

707 Based on these findings, several authors have suggested that the impact of
708 technologies in education depends on various variables and not on the tech-
709 nology itself, such as context, teaching and professional development practices,
710 curriculum of study and learner characteristics (Underwood, 2009; Tamim, 2011;
711 Higgins et al., 2012; Archer and Savage, 2014; Sung et al., 2016; Haßler, Major,
712 and Hennessy, 2016; Chauhan, 2017; Lee et al., 2020).

713 While the impact of ICT on student achievement has been studied in depth
714 by researchers, other aspects related to school life that are also affected by ICT,
715 such as equality, inclusion and social integration, have received less attention.
716 Further analysis of the literature review revealed greater investment in ICT in-
717 terventions to support learning and teaching in the core subjects of literacy,

718 mathematics and science. These were the most commonly studied topics in the
719 articles reviewed, often relying on national test scores, while studies of other
720 areas, such as social studies, were limited (Condie and Munro, 2007; Chauhan,
721 2017). That said, research still lacks impact studies that focus on the effects of
722 ICT on a range of curriculum subjects. Qualitative research provided additional
723 insights into the impact of digital technologies in education, documenting pos-
724 itive effects and providing more detail on implications, recommendations and
725 future research directions.

726 More specifically, the results regarding the role of ICT in supporting learn-
727 ing highlight the importance of teachers' pedagogical practice and the learning
728 context in the use of technologies and, therefore, their impact on teaching .

729 Schmid et al., 2014; Tamim et al., 2015).

730 The review also provided useful insights into the various factors that influ-
731 ence the impact of digital technologies in education. These factors are intercon-
732 nected and play a vital role in the transformation process. Specifically, these
733 factors include: a) digital skills, b) personal characteristics and professional de-
734 velopment of teachers, c) school leadership and management, d) connectivity,
735 infrastructure and government support, e) administration and data manage-
736 ment practices, f) students' socio-economic context and family support, and g)
737 socio-economic school context and emergency situations. It should be noted
738 that we observed factors that influence the integration of ICT in education, but
739 which can also be affected by it. For example, students' frequent use of ICT
740 and use of laptops for educational purposes positively affect the development of
741 digital skills (Zheng et al., 2016) and at the same time, digital skills affect the
742 use of ICT (Higgins et al., 2012; Fu, 2013).

743 5. Conclusions

744 Digital technologies offer several opportunities for fundamental improve-
745 ments in schools. However, investments in ICT infrastructure and professional
746 development to improve school education have not yet yielded fruitful results.

747 Digital transformation is a complex process that requires large-scale transforma-
748 tive changes that presuppose digital capability and readiness. To achieve such
749 changes, all stakeholders in the school ecosystem must share a common vision
750 regarding the integration of ICT in education and work towards achieving this
751 goal. In this work, we have provided useful information on the impacts of ICT
752 on different stakeholders in schools. Additionally, we have opened a forum for
753 discussion on the factors that affect the digital capacity and transformation of
754 schools. We hope that our study will inform policy, practice and research and
755 lead to a paradigm shift towards more holistic approaches in impact study and
756 evaluation.

757 **6. Study limitations and future directions**

758 The study results were based on a non-systematic literature review based on
759 the acquisition of documentation in specific databases. Future studies should
760 investigate several other databases to corroborate and improve our results. Ad-
761 ditionally, search queries could be enhanced with key terms that could provide
762 additional information on ICT integration in education, such as policies and
763 strategies for ICT integration in education.

764 Furthermore, the study relied on meta-analyses and literature reviews to
765 acquire evidence on the effects of ICT integration in schools. The nature of
766 the above studies is different as they are based on different research method-
767 ologies and data collection processes. Even in similar methodological studies
768 (e.g., meta-analyses), impact has been measured differently, depending on poli-
769 cies or research objectives (e.g., national exam results, pre/post testing). Fur-
770 thermore, qualitative studies provided additional insights based on self-reports
771 and research opinions on several different aspects and stakeholders that could
772 affect and be affected by ICT integration. As a result, it was difficult to es-
773 tablish causal relationships between so many interrelated variables. Despite
774 the challenges mentioned above, this study planned to examine academic units
775 as ecosystems consisting of multiple actors by bringing together several vari-

ables from different research epistemologies to provide an understanding of ICT integration. We propose that future studies aim for similar investigations by applying more holistic approaches to impact evaluation.

7. How to Prepare to Digital Transformation

- Converging Trends Drive Digital Transformation.
- Create Your Seven-Step Digital Transformation Plan.
- Define Objectives.
- Create Core Team.
- Analyze and Engage Stakeholders.
- Develop Digital Roadmap.
- Create Implementation Plan.
- Execute and Monitor.

How to Create a Digital Transformation Strategy and Plan

Soren Kaplan

Soren Kaplan

Bestselling Author & Speaker, Strategy & Innovation Consulting, Inc. Magazine, Praxie.com, Center for Effective Organizations at USC

9 février 2024

Digital transformation is no longer a buzzword. It's a critical business necessity. By creating a digital organization, you can remain relevant and competitive. It's tough to remain effective in business without going digital.

According to the research website Exploding Topics, 77 percent of companies have already started their digital transformation journey. The good news is that most organizations have embraced this critical imperative.

The bad news: Only 35 percent of organizations are successful in their digital transformation efforts.

Converging Trends Drive Digital Transformation

The convergence of trends and technologies that includes the cloud, artificial intelligence, the Internet of Things, business process automation, and other ad-

805 vancements have led to dramatic change. Not just in the technologies powering
806 business, but in the cultures that underlie how successful organizations operate
807 on a day-to-day basis.

808 True transformation is a journey. It requires a clear understanding of the
809 market dynamics impacting your world, of your business objectives, and of
810 the unique value you want to digitally deliver. It involves innovating business
811 processes as well as business models. It's about creating new levels of efficiencies,
812 customer experiences, and go-to-market strategies.

813 Take Microsoft's shift under CEO Satya Nadella, for example. The tech
814 giant has successfully transitioned from a software-centric business into a cloud-
815 based, service-based business model and organization. Microsoft set clear dig-
816 ital goals, assembled a dedicated cross-functional team, engaged stakeholders,
817 and developed a robust digital strategy focused on Azure and Office 365. The
818 implementation was carefully executed in phases, with continuous monitoring
819 and adaptation to market responses and technological trends. Microsoft's shift
820 wasn't just technological but also cultural, requiring a deep-seated change in
821 mindset across the organization.

822 Another example of digital transformation in the manufacturing sector is
823 General Electric. GE revolutionized its manufacturing by building its own dig-
824 ital platform, Predix, to predict when machines need repair before they break
825 down. This smart system focused on preventive maintenance, and, combined
826 with a virtual model that simulates a manufacturing line called a Digital Twin,
827 helps GE make and improve products by running simulations. By focusing on
828 digital software as a core capability, GE has made its factories smarter, pre-
829 venting problems and saving time and money, changing the way the company
830 creates and maintains its products.

831 Create Your Seven-Step Digital Transformation Plan

832 Digital innovators who want to tackle transformation don't have to boil the
833 ocean. The best transformations, whether digital or not, focus on quick wins
834 and early adopters to build momentum. While a bigger-picture strategy can be
835 helpful, just getting started is what's most important.

836 Here's a seven-step approach for not just getting going with your digital
837 transformation, but ensuring it's successful. It's based on my 30 years of strat-
838 egy, change management, and culture transformation experience across dozens
839 of Fortune 1000 companies, as well as mid-size organizations. Each step includes
840 a core strategy supported by the three most critical key actions to take to drive
841 the strategy successfully:

842 1. Define Objectives

843 Establish clear, measurable goals for your digital transformation that align
844 with your organization's overall strategy.

845 Key actions:

846 Conduct workshops with senior leadership to identify strategic objectives

847 Translate these objectives into specific digital goals (e.g., enhancing customer
848 experience, improving operational efficiency)

849 Develop KPIs to measure the success of the digital transformation

850 2. Create Core Team

851 Assemble a cross-functional team with members from different departments
852 to drive the transformation.

853 Key actions:

854 Identify individuals with diverse skills (IT, operations, HR, etc.) and a
855 digital mindset

856 Assign clear roles and responsibilities to each team member

857 Establish regular team meetings for progress updates and issue resolution

858 3. Analyze and Engage Stakeholders

859 Identify and understand the stakeholders impacted by the digital transfor-
860 mation and strategize their engagement.

861 Key actions:

862 Map stakeholders based on their influence and interest in the project

863 Develop tailored communication plans for each stakeholder group

864 Schedule regular updates and feedback sessions with key stakeholders

865 4. Develop Digital Roadmap

866 Formulate a phased strategy that outlines the approach, technologies to be
867 adopted, and implementation roadmap.

868 Key actions:

869 Evaluate current digital capabilities and identify gaps

870 Define optimal digital processes and applications

871 Create a phased roadmap with clear milestones and deliverables

872 5. Create Implementation Plan

873 Develop detailed plans for the execution of the digital processes and appli-
874 cations.

875 Key actions:

876 Define project scopes, timelines, and resource requirements for each initiative

877 Develop risk management plans to address potential challenges

878 Establish a governance model to oversee the implementation process

879 6. Execute and Monitor

880 Implement the transformation initiatives and continuously monitor progress
881 against milestones and metrics.

882 Key actions:

883 Execute initiatives according to the implementation plan

884 Monitor progress using the defined KPIs and adjust strategies as needed

885 Conduct regular team meetings to review progress and resolve issues

886 7. Track and Optimize

887 Assess progress, identify areas for ongoing improvement, and iterate imple-
888 mentation approaches to ensure agility.

889 Key actions:

890 Gather feedback from stakeholders and analyze performance data

891 Identify successes and areas needing improvement

892 Develop a plan for continuous optimization and adaptation

893 In today's world, if you're not digitally innovating, you're likely declining as
894 an organization. Use these strategies to stay ahead.

895 **8. Keys to Digital Transformation**

896 Align on the Why of Digital Transformation.

897 Prepare for Culture Change.

898 Start Small But Strategic.

899 Map Out Technology Implementation.

900 Seek Out Partners and Expertise.

901 Gather Feedback and Refine as Needed.

902 Scale and Transform.



Figure 2: Graphical abstract

903 **9. Steps to a Successful Digital Transformation Strategy**

904 **10. Unexpected Ways Digital Transformation Benefits**

905 **11. Unexpected Ways Digital Transformation Benefits**

906 This study examines smart education in economic monitoring and crisis pre-
907 vention, recognizing that education plays a leading role in awareness raising,
908 crisis mitigation and general human security in the sustainability. It presents a
909 multi-modal deep learning model based on CNN-LSTM while discussing the in-
910 tegration of artificial intelligence (AI), Internet of Things (IoT), big data, smart
911 data, of supercomputing and social networks in economic surveillance and crisis
912 prevention through intelligent education. This multimodal model improves eco-
913 nomic activity by providing interactive and intelligent education to alert people,
914 especially managers, of an ongoing event in order to make quick decisions. The
915 study has a solid theoretical basis and demonstrates rigor and empirical valida-
916 tion. LSTM learns long-term dependencies: it takes the concatenated function
917 of business, marketing or financial data as input while integrating encapsula-
918 tions, using wearables, smart data and social networks to capture relevant secure
919 information with CNN in combining various research results. He is shown to
920 learn to locate and recognize several aspects of marketing, business and finance.
921 The model is evaluated on the challenging tasks of the COVID-19 health crisis,
922 electrical fraud detection in business and finance, and counterfeiting: it is shown
923 to be more accurate than state-of-the-art art of other neural networks. and uses
924 less training time.

925 Foresight: the key to a successful digital transformation OPTIMA TEAM
926 SEPTEMBER 28, 2022

927 Much has been written about digital transformation, the new industrial rev-
928 olution driven by rapid advances in new technologies. Digitization, digitization,
929 paperless processes, automation - these are the so-called projects in every com-
930 pany. Certainly, it offers opportunities to push the boundaries in many areas.
931 However, leading a digital transformation that delivers on this promise is not
932 easy, as not every project will lead to success.

933 The complexity lies in the execution, because pushing the boundaries of what
934 is possible, even if technologies allow it, can only be achieved by changing the
935 way we think today. This means that we must be ready to revise our processes
936 according to new technological paradigms. This is the reason why most digital
937 transformations remain unsuccessful despite the new technologies available and
938 accessible today.

939 To be or not to be “Customer-centric”? That is the question We have gotten
940 into the habit of always and everywhere being “customer-centric”, that is, you
941 should put yourself in the customer’s place, how they think and live their life.
942 Think about solutions that will solve their daily problems.

943 Customer orientation is definitely the right approach to guarantee a long-
944 term business. Knowing the needs and expectations of customers allows you to
945 run a proven business and survive in an environment where everyone takes a cer-
946 tain place according to the known rules of the game in order to continue growing
947 and extract maximum profitability from a business model. given company.

948 As counterintuitive as it may seem, customer centricity may not be the right
949 path for digital transformation.

950 Changing the existing rituals, habits, established logics, and frictions that
951 have become part of the norm, to create new foundations, new interactions, new
952 applications, and new processes that create a new generation of value proposi-
953 tions, you may need to -be thinking differently. More than thinking the same
954 way about your customers, anticipate them and lead them to their future needs.

955 As Henry Ford said: “If I had asked people what they wanted, they would
956 have said: faster horses.”

957 Indeed, customers of your services can easily describe and express their
958 needs, but only within the limits of what they see and know about the present.
959 They can hardly do this if they exceed this limit. So, the first thing you need to
960 do to achieve a successful digital transformation is not to be customer-centric
961 but to look forward to the real needs of your customers.

962 Today is the present and tomorrow should not be a mystery Defining the
963 needs of your customers also means characterizing the present. How you segment

964 them is how you look at them and how you analyze the existing order as it is
965 today. It's important to position your customers and then move within that
966 order.

967 A digital transformation means that we must break with the way we perceive
968 them, with the dynamics, with the segmentation and even with the logic of their
969 own evolution.

970 The digital transformation approach removes segmenting your customers as
971 a prerequisite. While some look at expectations and requirements separately,
972 such as easy access to raw data, insightful reporting to aid decision-making, or
973 compliance with market regulations, implementing digital transformation relies
974 on examining groups of needs that transcend customers and ultimately connect
975 them into a single, homogeneous demand.

976 How can all this be applied to the digitalization of laboratory processes?
977 A laboratory has several types of customers: internal and external, to say the
978 least. If you're leading a digital transformation in a lab, you also need to view
979 stakeholders and users as customers. They all have needs, requirements and
980 expectations.

981 The technology implemented must serve multiple functions within the or-
982 ganization. It should also make processes and end users' lives easier, and yes,
983 it should also have a positive impact on the customer experience and reinforce
984 their willingness to continue doing business with you.

985 When a laboratory is considering adopting new technology to support its re-
986 source management, this could come from a concrete phase of the organization's
987 digital transformation strategy. Perhaps the laboratory is already experiencing
988 the revolution or is simply wondering where to start.

989 Digitalization of resource management has immediate benefits, as it channels
990 and coordinates all related information on task planning, staff skills, routine
991 instrument maintenance and status overview of projects. This makes laboratory
992 productivity and efficiency immediately visible.

993 Whoever uses this tool benefits from access to the precise information they
994 need for their decision-making and daily work.

995 Technology allows you to define the rights and access of your users so that
996 they only receive the information they need to access. And that’s where the
997 customer-centric approach comes in. What do your customers need today, and
998 what can we enable them to do today so we can anticipate their needs tomorrow?

999 For a “simple” tool like a laboratory resource management system, it’s just
1000 a matter of configuration.

1001 Optima does not need to be customized and requires no coding; it can be
1002 configured and adapted to the needs of your users.

1003 There are settings and configuration settings that can be set as default during
1004 the implementation phase. Further settings for reporting activities and data
1005 visualization can be set by end users themselves based on their current role and
1006 needs.

1007 A laboratory scheduling system is essential to meet the real needs of your
1008 users. It allows the laboratory to keep activities and procedures under control,
1009 ensuring efficiency and performance.

1010 But it doesn’t stop there.

1011 As we’ve highlighted in previous sections of this article, the way to success-
1012 fully implement a digital transformation is to think about the future and not
1013 just the needs of today’s customers.

1014 With this in mind, we can strongly state that a laboratory planning system
1015 is the right tool that can help laboratory managers, and more generally the
1016 management of a company, develop a long-term vision: when productivity and
1017 laboratory efficiency are immediately visible, it definitely becomes easier, even
1018 obvious, which path to take for future and lasting success.

1019 12. Conclusion and future works

1020 In conclusion, using a CNN-LSTM-based hybrid model for health monitoring
1021 and health crisis forecasting has the potential to fundamentally change how
1022 people, especially managers, improve their tasks of daily life without worry
1023 or fear, examining how IoT, smart data, and multiple social media sources

1024 technologies are integrated.

1025 The results provide insight into the perceived impact of this educational
1026 awareness on alertness, effectiveness, happiness, and favourable attitudes.

1027 According to the study, CNN-LSTM-based hybrid systems provide a real-
1028 istic and safe learning environment and show how they improve health crisis
1029 forecasting.

1030 Public health officials can be confident that their fellow citizens live in peace,
1031 knowing that they are in a regulated and secure environment where life is safe
1032 which helps them trust their health officials and live with confidence.

1033 The study results are consistent with the idea that this new hybrid ap-
1034 proachto health monitoring and health crisis forecasting systems can bridge
1035 the gap between theoretical understanding and real-world application, thereby
1036 producing better qualified professionals in a regulated and safe environment.

1037 The study proposes a novel approach to monitoring the health environment
1038 using educational awareness and safe information. To avoid false information,
1039 it first uses multimodal features namely connected objects, smart data and/or
1040 multiple social media sources which allows to check if the content is not safe.
1041 Then, if this fails, we have the context-sentiment-based hybrid model to avoid
1042 unsafe information.

1043 However, the hypothesis that "digital health interventions can increase re-
1044 sponse capacity to COVID-19" [77] can be considered susceptible to adversarial
1045 attacks, domain-specific biases, or unseen data (false information).

1046 Fortunately, a technological ecosystem integrating different information sys-
1047 tems was developed to meet the needs highlighted in the international regula-
1048 tions governing the response to the pandemic [77]. In this way, false information
1049 was avoided.

1050 In future work, we plan to include a comparative analysis of this model
1051 using several state-of-the-art approaches to fake news detection simultaneously
1052 to provide researchers and practitioners with a better understanding of the
1053 strengths of our approach.

1054 Bert, the natural language processing tool, is extended to recognize the

1055 context and sentiment of the news, using the visual mechanism.

1056 This new model, suitable for temporal dimension analysis, is applied to
1057 prevent false news. The context and the sentiment are incorporated as very
1058 important characteristics to enhance the accuracy. The performance of the new
1059 method is studied using a public dataset. The results show that this new hybrid
1060 model for preventing fake news from connected objects, smart data, and multiple
1061 social media sources is more efficient than other models (SVM and others).

1062 Advances in language modeling enable the automatic identification of mis-
1063 leading and deceptive news. The study focused on training deep learning models
1064 to extract patterns from false or real news. The study shows results for identify-
1065 ing fake and real news, but a major drawback of the model remains the need to
1066 continually update it to deal with the arrival of new false news that is constantly
1067 arriving.

1068 Although the study highlights the potential advantages of this new approach,
1069 it is important to know the people's opinions, in particular, feedback on the use
1070 of this new approach to health surveillance and crisis forecasting using descrip-
1071 tive statistics through sampling in order to validate this new system. This will
1072 be the task of future work.

1073 In the future, integrating IoT technologies (as sensors), smart data, and mul-
1074 tiple social media sources to capture relevant information using multimodal deep
1075 learning will enable the incorporation of multimodal features such as audio and
1076 video, the integration of external knowledge bases for fact-checking and lever-
1077 aging set learning techniques to enhance performance, relevance and research
1078 impact.

1079 Strengthen educational communication with online courses and educational
1080 awareness to warn and notably fight re-emerging epidemics such as syphilis
1081 and other sexually transmitted infections, cholera, polio, and all other global
1082 re-emerging epidemics to strengthen and promote health education.

1083 **Acknowledgments**

1084 We acknowledge the support of Direction Générale de la Recherche Scien-
1085 tifique et du Développement Technologique 'DGRSDT'. MESRS, Algeria.

1086 **Declarations**

1087 *Funding does not require any funding: this work is within the framework
1088 of research

1089 * Conflicts of interest: All metrics are the property of the LMA laboratories
1090 of the University of Bejaia, Algeria

1091 *Availability of data and material: The data supporting this study's find-
1092 ings are available from the corresponding author. However, restrictions apply
1093 to the availability of these data, which were used in the LMA laboratory from
1094 the University of Bejaia, in Algeria for the current study, and so are not pub-
1095 licly available. Data are however available from the corresponding author upon
1096 reasonable request and with permission of Managers of the LMA laboratory.

1097 **Author Agreement Statement**

1098 We the undersigned declare that this manuscript is original, has not been
1099 published before and is not currently being considered for publication elsewhere.

1100 We confirm that the manuscript has been read and approved by all named
1101 authors and that there are no other persons who satisfied the criteria for au-
1102 thorship but are not listed. We further confirm that the order of authors listed
1103 in the manuscript has been approved by all of us.

1104 We understand that the Corresponding Author is the sole contact for the
1105 Editorial process.

1106 He is responsible for communicating with the other authors about progress,
1107 submissions of revisions and final approval of proofs

1108 Signed by all authors as follows:

1109 Z. Bouzidi, and

1110 A.M. Boudries

1111 References

- 1112 [1] Archer, K., Savage, R., Sanghera-Sidhu, S., Wood, E., Gottardo, A., &
 1113 Chen, V. (2014). Examining the effectiveness of technology use in class-
 1114 rooms: A tertiary meta-analysis. *Computers & Education*, 78, 140-149.
 1115 <https://doi.org/10.1016/j.compedu.2014.06.001>

- 1116 [2] Balanskat, A. (2006). The ICT Impact Report: A review of
 1117 studies of ICT impact on schools in Europe, European School-
 1118 net. Retrieved from: [https://en.unesco.org/icted/content/ict-impact-](https://en.unesco.org/icted/content/ict-impact-reportreview-studies-ict-impact-schools-europe)
 1119 [reportreview-studies-ict-impact-schools-europe](https://en.unesco.org/icted/content/ict-impact-reportreview-studies-ict-impact-schools-europe)

- 1120 [3] Balanskat, A., Blamire, R., & Kefala, S. (2006). The
 1121 ICT impact report. European Schoolnet. Retrieved from:
 1122 http://colceti.colfinder.org/sites/default/files/ict_impact_report_0.pdf

- 1123 [4] Balanskat, A. (2009). Study of the impact of technology in primary schools
 1124 - Synthesis Report. Empirica and European Schoolnet. Retrieved from:
 1125 https://erte.dge.mec.pt/sites/default/files/Recursos/Estudos/synthesis_report_steps_en.pdf

- 1126 [5] Balyer, A., & Öz, Ö. (2018). Academicians' views on dig-
 1127 ital transformation in education. *International Online Jour-*
 1128 *nal of Education and Teaching (IOJET)*, 5(4), 809-830.
 1129 <http://iojet.org/index.php/IOJET/article/view/441/295>

- 1130 [6] Bates, A. W. (2015). Teaching in a digital age: Guidelines for designing
 1131 teaching and learning. Open Educational Resources Collection. 6. Retrieved
 1132 from: <https://irl.umsl.edu/oer/6>

- 1133 [7] Bingimlas, K. A. (2009). Barriers to the successful integration of ICT in
 1134 teaching and learning environments: A review of the literature. *Eurasia*
 1135 *Journal of Mathematics, science and technology education*, 5(3), 235-245.
 1136 <https://doi.org/10.12973/ejmste/75275>

- 1137 [8] Brooks, D. C. & McCormack, M. (2020). Driving Digital Transformation
1138 in Higher Education. Retrieved from: [https://library.educause.edu/-](https://library.educause.edu/-/media/files/library/2020/6/dx2020.pdf?la=en&hash=28FB8C377B59AFB1855C225BBA8E3CFBB0A27)
1139 [/media/files/library/2020/6/dx2020.pdf?la=en&hash=28FB8C377B59AFB1855C225BBA8E3CFBB0A27](https://library.educause.edu/-/media/files/library/2020/6/dx2020.pdf?la=en&hash=28FB8C377B59AFB1855C225BBA8E3CFBB0A27)
- 1140 [9] Chauhan, S. (2017). A meta-analysis of the impact of technology on learn-
1141 ing effectiveness of elementary students. *Computers & Education*, 105, 14-
1142 30. <https://doi.org/10.1016/j.compedu.2016.11.005>
- 1143 [10] Cheok, M. L., & Wong, S. L. (2015). Predictors of e-learning satisfaction in
1144 teaching and learning for school teachers: A literature review. *International*
1145 *Journal of Instruction*, 8(1), 75-90.
- 1146 [11] Cheung, A. C., & Slavin, R. E. (2011). The Effectiveness of Education
1147 Technology for Enhancing Reading Achievement: A Meta-Analysis. Center
1148 for Research and reform in Education.
- 1149 [12] Condie, R., & Munro, R. K. (2007). The impact
1150 of ICT in schools-a landscape review. Retrieved from:
1151 https://oei.org.ar/ibertic/evaluacion/sites/default/files/biblioteca/33_impact_ict_in_schools.pdf
- 1152 [13] Conrads, J., Rasmussen, M., Winters, N., Geniet, A., Langer, L., Baci-
1153 galupo, M., & Punie, Y. (2017). Digital education policies in Europe and
1154 beyond. JRC Science for policy report.
- 1155 [14] Costa, P., Castaño-Muñoz, J., & Kamylylis, P. (2021). Cap-
1156 turing schools' digital capacity: Psychometric analyses of the
1157 SELFIE self-reflection tool. *Computers & Education*, 162, 104080.
1158 <https://doi.org/10.1016/j.compedu.2020.104080>
- 1159 [15] Cussó-Calabuig, R., Farran, X. C., & Bosch-Capblanch, X. (2018). Effects
1160 of intensive use of computers in secondary school on gender differences in
1161 attitudes towards ICT: A systematic review. *Education and Information*
1162 *Technologies*, 23(5), 2111-2139. [https://doi.org/10.1007/s10639-018-9706-](https://doi.org/10.1007/s10639-018-9706-6)
1163 6

- [16] Daniel, S. J. (2020). Education and the COVID-19 pandemic. *Prospects*, 49(1), 91-96.
- [17] Delcker, J., & Ifenthaler, D. (2021). Teachers' perspective on school development at German vocational schools during the Covid-19 pandemic. *Technology, Pedagogy and Education*, 30(1), 125-139. <https://doi.org/10.1080/1475939X.2020.1857826>
- [18] Delgado, A. J., Wardlow, L., McKnight, K., & O'Malley, K. (2015). Educational technology: A review of the integration, resources, and effectiveness of technology in K-12 classrooms. *Journal of Information Technology Education*, 14. <http://www.jite.org/documents/Vol14/JITEv14ResearchP397-416Delgado1829.pdf>
- [19] Di Pietro, G., Biagi, F., Costa, P., Karpiński, Z., & Mazza, J. (2020). The likely impact of COVID-19 on education: Reflections based on the existing literature and recent international datasets (Vol. 30275). Luxembourg: Publications Office of the European Union.
- [20] Elkordy, A., & Lovinelli, J (2020). Competencies, Culture, and Change: A Model for Digital Transformation in K12 Educational Contexts. In D. Ifenthaler, S. Hofhues, M. Egloffstein & C. Helbig (Eds). *Digital Transformation of Learning Organizations* (pp. 203-219). Springer.
- [21] Eng, T. S. (2005). The impact of ICT on learning: A review of research. *International Education Journal*, 6(5), 635-650.
- [22] European Commission. (2020). Digital Education Action Plan 2021 – 2027. Resetting education and training for the digital age. https://ec.europa.eu/education/sites/default/files/documentlibrary-docs/deap-communication-sept2020_en.pdf
- [23] European Commission. (2019). 2nd survey of schools: ICT in education. Objective 1: Benchmark progress in ICT in schools.

- 1191 Retrieved from: [https://data.europa.eu/euodp/data/storage/f/2019-03-](https://data.europa.eu/euodp/data/storage/f/2019-03-19T084831/FinalreportObjective1-BenchmarkprogressinICTinschools.pdf)
1192 [19T084831/FinalreportObjective1-BenchmarkprogressinICTinschools.pdf](https://data.europa.eu/euodp/data/storage/f/2019-03-19T084831/FinalreportObjective1-BenchmarkprogressinICTinschools.pdf)
- 1193 [24] Eurydice (2019). Digital Education at School in Europe, Luxem-
1194 bourg: Publications Office of the European Union. Retrieved from:
1195 [https://eacea.ec.europa.eu/nationalpolicies/eurydice/content/digital-](https://eacea.ec.europa.eu/nationalpolicies/eurydice/content/digital-education-school-europe_en)
1196 [education-school-europe_en](https://eacea.ec.europa.eu/nationalpolicies/eurydice/content/digital-education-school-europe_en)
- 1197 [25] Escueta, M., Quan, V., Nickow, A. J., & Oreopoulos, P.
1198 (2017). Education technology: An evidence-based review.
1199 <https://ssrn.com/abstract=3031695>
- 1200 [26] Fernández-Gutiérrez, M., Gimenez, G., & Calero, J. (2020). Is the use of
1201 ICT in education leading to higher student outcomes? Analysis from the
1202 Spanish Autonomous Communities. *Computers & Education*, 157, 103969.
1203 <https://doi.org/10.1016/j.compedu.2020.103969>
- 1204 [27] Friedel, H., Bos, B., Lee, K., & Smith, S. (2013, March). The impact of mo-
1205 bile handheld digital devices on student learning: A literature review with
1206 meta-analysis. In *Society for Information Technology & Teacher Education*
1207 *International Conference* (pp. 3708-3717). Association for the Advancement
1208 of Computing in Education (AACE).
- 1209 [28] Fu, J. S. (2013). ICT in Education: A Critical Literature Review and Its
1210 Implications. *International Journal of Education and Development using*
1211 *Information and Communication Technology (IJEDICT)*, 9 (1), 112-125.
- 1212 [29] Gaol, F. L., & Prasolova-Førland, E. (2021). Special section editorial: The
1213 frontiers of augmented and mixed reality in all levels of education. *Educa-*
1214 *tion and Information Technologies*, 1-13.
- 1215 [30] Garzón, J., Baldiris, S., Gutiérrez, J., & Pavón, J. (2020). How do peda-
1216 gogical approaches affect the impact of augmented reality on education?
1217 A meta-analysis and research synthesis. *Educational Research Review*,
1218 100334. <https://doi.org/10.1016/j.edurev.2020.100334>

- [31] Garzón, J., & Acevedo, J. (2019). Meta-analysis of the impact of Augmented Reality on students' learning gains. *Educational Research Review*, 27, 244-260. <https://doi.org/10.1016/j.edurev.2019.04.001>
- [32] Grgurović, M., Chapelle, C. A., & Shelley, M. C. (2013). A meta-analysis of effectiveness studies on computer technology-supported language learning. *ReCALL*, 25(2), 165-198. <https://doi.org/10.1017/S0958344013000013>
- [33] Haßler, B., Major, L., & Hennessy, S. (2016). Tablet use in schools: A critical review of the evidence for learning outcomes. *Journal of Computer Assisted Learning*, 32(2), 139-156. <https://doi.org/10.1111/jcal.12123>
- [34] Hardman, J. (2019). Towards a pedagogical model of teaching with ICTs for mathematics attainment in primary school: A review of studies 2008–2018. *Heliyon*, 5(5), e01726. <https://doi.org/10.1016/j.heliyon.2019.e01726>
- [35] Hattie, J., Rogers, H. J., & Swaminathan, H. (2014). The role of meta-analysis in educational research. In A. D. Reid, P. Hart, & M.A. Peters (Eds.) *A companion to research in education* (pp. 197-207). Springer, Dordrecht.
- [36] Hattie, J. (2008). *Visible learning: A synthesis of over 800 meta-analyses relating to achievement*. Routledge. <https://doi.org/10.4324/9780203887332>
- [37] Higgins, S., Xiao, Z., & Katsipatakis, M. (2012). *The impact of digital technology on learning: A summary for the education endowment foundation*. Durham, UK: Education Endowment Foundation and Durham University.
- [38] Hillmayr, D., Ziernwald, L., Reinhold, F., Hofer, S. I., & Reiss, K. M. (2020). The potential of digital tools to enhance mathematics and science learning in secondary schools: A context-specific meta-analysis. *Computers & Education*, 153, 1038–97. <https://doi.org/10.1016/j.compedu.2020.103897>
- [39] Huang, R., Ritzhaupt, A. D., Sommer, M., Zhu, J., Stephen, A., Valle, N., ... & Li, J. (2020). The impact of gamification in educational settings on stu-

- 1247 dent learning outcomes: a metaanalysis. *Educational Technology Research*
1248 and Development, 68(4), 1875-1901. [https://doi.org/10.1007/s11423-020-](https://doi.org/10.1007/s11423-020-09807-z)
1249 09807-z
- 1250 [40] Istenic Starcic, A., & Bagon, S. (2014). ICT- supported learning for inclu-
1251 sion of people with special needs: Review of seven educational technology
1252 journals, 1970–2011. *British Journal of Educational Technology*, 45(2), 202-
1253 230. <https://doi.org/10.1111/bjet.12086>
- 1254 [41] Jewitt, C., Clark, W., & Hadjithoma-Garstka, C. (2011). The use
1255 of learning platforms to organise learning in English primary and
1256 secondary schools. *Learning, Media and Technology*, 36(4), 335-348.
1257 <https://doi.org/10.1080/17439884.2011.621955>
- 1258 [42] JISC (2020). What is digital transformation?. Retrieved from:
1259 [https://www.jisc.ac.uk/guides/digital-strategy-framework-for-university-](https://www.jisc.ac.uk/guides/digital-strategy-framework-for-university-leaders/what-is-digitaltransformation)
1260 leaders/what-is-digitaltransformation
- 1261 [43] Kao, Chian-Wen (2014). The effects of digital game-based learning task in
1262 English as a foreign language contexts: A meta-analysis. *Education Jour-*
1263 *nal*, 42(2), 113-141
- 1264 [44] König, J., Jäger-Biela, D. J., & Glutsch, N. (2020). Adapting to
1265 online teaching during COVID-19 school closure: teacher educa-
1266 tion and teacher competence effects among early career teachers in
1267 Germany. *European Journal of Teacher Education*, 43(4), 608-622.
1268 <https://doi.org/10.1080/02619768.2020.1809650>
- 1269 [45] Lawrence, J. E. & Tar, U. A. (2018) Factors that influ-
1270 ence teachers' adoption and integration of ICT in teach-
1271 ing/learning process, *Educational Media International*, 55(1), 79-
1272 105<https://doi.org/10.1080/09523987.2018.1439712>
- 1273 [46] Lee, S., Kuo, L. J., Xu, Z., & Hu, X. (2020). The effects of technology-
1274 integrated classroom instruction on K-12 English language learners' literacy

- development: a meta-analysis. *Computer Assisted Language Learning*, 1-32. <https://doi.org/10.1080/09588221.2020.1774612>
- [47] Liao, Y. K. C., Chang, H. W., & Chen, Y. W. (2007). Effects of Computer Application on Elementary School Student's Achievement: A Meta-Analysis of Students in Taiwan. *Computers in the Schools*, 24(3-4), 43-64. https://doi.org/10.1300/J025v24n03_04
- [48] Li, Q., & Ma, X. (2010). A meta-analysis of the effects of computer technology on school students' mathematics learning. *Educational Psychology Review*, 22(3), 215-243.
- [49] OECD (2015). *Students, Computers and Learning: Making the Connection*. PISA, OECD Publishing, Paris. Retrieved from: <https://doi.org/10.1787/9789264239555-en>
- [50] OECD (2021). *OECD Digital Education Outlook 2021: Pushing the Frontiers with Artificial Intelligence, Blockchain and Robots*. Retrieved from: https://www.oecd-ilibrary.org/education/oecd-digital-education-outlook-2021_589b283f-en
- [51] Pettersson, F. (2021). Understanding digitalization and educational change in school by means of activity theory and the levels of learning concept. *Education and Information Technologies*, 26(1), 187-204.
- [52] Pihir, I., Tomičić-Pupek, K., & Furjan, M. T. (2018). Digital transformation insights and trends. In *Central European Conference on Information and Intelligent Systems* (pp. 141-149). Faculty of Organization and Informatics Varazdin. Retrieved from <https://www.proquest.com/conference-papers-proceedings/digital-transformation-insightstrends/docview/2125639934/se-2>
- [53] Punie, Y., Zinnbauer, D., & Cabrera, M. (2006). A review of the impact of ICT on learning Working Paper prepared for DG EAC. Retrieved from: http://www.eurosfair.prd.fr/7pc/doc/1224678677_jrc47246n.pdf

- 1303 [54] Redep, N. B. (2021). Comparative overview of the digital preparedness of
1304 education systems in selected CEE countries. Center for Policy Studies.
1305 CEU Democracy Institute.
- 1306 [55] Rott, B., & Marouane, C. (2018). Digitalization in schools—organization,
1307 collaboration and communication. In *Digital Marketplaces Unleashed* (pp.
1308 113-124). Springer, Berlin, Heidelberg.
- 1309 [56] Schmid, R. F., Bernard, R. M., Borokhovski, E., Tamim, R. M.,
1310 Abrami, P. C., Surkes, M. A., Wade, C., A., & Woods, J. (2014).
1311 The effects of technology use in postsecondary education: A meta-
1312 analysis of classroom applications. *Computers & Education*, 72, 271-291.
1313 <https://doi.org/10.1016/j.compedu.2013.11.002>
- 1314 [57] Schuele, C. M., & Justice, L. M. (2006). The importance of effect sizes
1315 in the interpretation of research: Primer on research: Part 3. *The ASHA*
1316 *Leader*, 11(10), 14-27. <https://doi.org/10.1044/leader.FTR4.11102006.14>
- 1317 [58] Sellar, S. (2015). Data infrastructure: a review of expanding ac-
1318 countability systems and largescale assessments in education, *Dis-*
1319 *course: Studies in the Cultural Politics of Education*, 36:5, 765-777
1320 <https://doi.org/10.1080/01596306.2014.931117>
- 1321 [59] Stock, W. A. (1994). Systematic coding for research synthesis. In H. Cooper
1322 & L. V. Hedges (Eds.) *The handbook of research synthesis*, 236, 125-138.
1323 New York: Russel Sage.
- 1324 [60] Sung, Y. T., Chang, K. E., & Liu, T. C. (2016). The effects of integrating
1325 mobile devices with teaching and learning on students' learning perfor-
1326 mance: A meta-analysis and research synthesis. *Computers & Education*,
1327 94, 252-275. <https://doi.org/10.1016/j.compedu.2015.11.008>
- 1328 [61] Talan, T., Doğan, Y., & Batdı, V. (2020). Efficiency of digital and non-
1329 digital educational games: A comparative meta-analysis and a meta-

- 1330 thematic analysis. *Journal of Research on Technology in Education*, 52(4),
1331 474-514. <https://doi.org/10.1080/15391523.2020.1743798>
- 1332 [62] Tamim, R. M., Borokhovski, E., Pickup, D., Bernard, R. M., &
1333 El Saadi, L. (2015). Tablets for teaching and learning: A system-
1334 atic review and meta-analysis. *Commonwealth of Learning*. Retrieved
1335 from: [http://oasis.col.org/bitstream/handle/11599/1012/2015_Tamim-et-](http://oasis.col.org/bitstream/handle/11599/1012/2015_Tamim-et-al_Tablets-forTeaching-and-Learning.pdf)
1336 [al_Tablets-forTeaching-and-Learning.pdf](http://oasis.col.org/bitstream/handle/11599/1012/2015_Tamim-et-al_Tablets-forTeaching-and-Learning.pdf)
- 1337 [63] Tamim, R. M., Bernard, R. M., Borokhovski, E., Abrami, P. C., & Schmid,
1338 R. F. (2011). What forty years of research says about the impact of technol-
1339 ogy on learning: A second-order metaanalysis and validation study. *Review*
1340 *of Educational research*, 81(1), 4-28. <https://doi.org/10.3102>
- 1341 [64] Tolani-Brown, N., McCormac, M., & Zimmermann, R. (2011). An analysis
1342 of the research and impact of ICT in education in developing country con-
1343 texts. In *ICTs and sustainable solutions for the digital divide: Theory and*
1344 *perspectives* (pp. 218-242). IGI Global.
- 1345 [65] Trucano, M. (2005). *Knowledge Maps: ICTs in Ed-*
1346 *ucation*. Washington, DC: info Dev / World Bank.
1347 <https://files.eric.ed.gov/fulltext/ED496513.pdf>
- 1348 [66] Underwood, J. D. (2009). The impact of digital technology: A review of
1349 the evidence of the impact of digital technologies on formal education.
- 1350 [67] Verschaffel, L., Depaepe, F., & Mevarech, Z. (2019). Learning Mathemat-
1351 ics in metacognitively oriented ICT-Based learning environments: A sys-
1352 tematic review of the literature. *Education Research International*, 2019.
1353 <https://doi.org/10.1155/2019/3402035>
- 1354 [68] Voogt, J., Knezek, G., Cox, M., Knezek, D., & ten Brummelhuis, A. (2013).
1355 Under which conditions does ICT have a positive effect on teaching and
1356 learning? A call to action. *Journal of Computer Assisted Learning*, 29(1),
1357 4-14. <https://doi.org/10.1111/j.1365-2729.2011.00453.x>

- [69] Vuorikari, R., Punie, Y., & Cabrera, M. (2020). Emerging technologies and the teaching profession: Ethical and pedagogical considerations based on near-future scenarios (No. JRC120183). Joint Research Centre. Retrieved from: <https://publications.jrc.ec.europa.eu/repository/handle/JRC120183>
- [70] Zheng, B., Warschauer, M., Lin, C. H., & Chang, C. (2016). Learning in one-to-one laptop environments: A meta-analysis and research synthesis. *Review of Educational Research*, 86(4), 1052-1084. <https://doi.org/10.3102>
- [71] Bouzidi Z., Amad M. & Boudries A., (2019), *Intelligent and Real-time Alert Model for Disaster Management based on Information retrieval from Multiple Sources*, *Journal of Advanced Media and Communication*, Vol. 7, No. 4, pp. 309-330, DOI: 10.1504/IJAMC.2019.111193
- [72] Bouzidi Z., Boudries A. & Amad M., (2020), *Towards a Smart Interface-based Automated Learning Environment Through Social Media for Disaster Management and Smart Disaster Education*, *Proceedings of Science and Information Conference - Advances in Intelligent Systems and Computing*, July 16th and 17th, 2020, (SAI 2020), vol. 1228, pp. 443-468, *Doi* : 10.1007/978 – 3 – 030 – 52249 – 0_31
- [73] K. Shankar, Eswaran Perumal, Mohamed Elhoseny, Fatma Taher, B. B. Gupta, Ahmed A. Abdelatif, (2022), Synergic Deep Learning for Smart Health Diagnosis of COVID-19 for Connected Living and Smart Cities, *ACM Transactions on Internet Technology* 22(3):1-14, DOI:10.1145/3453168
- [74] Mohammed Sabr, Brahami Menaouer, Abid Faten Fatima Zohra, Matta Nada, (2022), Sentiment Analysis of COVID Tweets Using Adaptive Neuro Fuzzy Inference System, *International Journal of Software Science and Computational Intelligence* 14(1):1-20, DOI: 10.4018/IJSSCI.300361
- [75] Janaína L. R. S. Valentim, Sara Dias-Trindade, Eloiza S. G. Oliveira, Manoel H. Romão, Felipe Fernandes, Alexandre R. Caitano, Marilyn A. A.

- 1386 Bonfim, Aline P. Dias, Cristine M. G. Gusmão, Philippi S. G. Morais,
1387 Ronaldo S. Melo, Gustavo Fontoura de Souza, Kelson C. Medeiros, Maria
1388 C. F. D. Rêgo, Ricardo B. Ceccim, and Ricardo A. M. Valentim, (2023),
1389 Evaluation of massive education in prison health: a perspective of health
1390 care for the person deprived of freedom in Brazil, Public Health Education
1391 and Promotion, Volume 11, <https://doi.org/10.3389/fpubh.2023.1239769>
- 1392 [76] Bouzidi Z., Amad M. & Boudries A., (2022), *Enhancing Warning, Sit-*
1393 *uational Awareness, Assessment and Education in Managing Emergency:*
1394 *Case study of Covid-19*, Springer Nature Computer Science, vol. 1, Number
1395 454, doi:10.1007/s42979-022-01351-2
- 1396 [77] Ricardo Alexsandro de Medeiros Valentim, Thaisa Santos Lima, Lyane
1397 Ramalho Cortez, Daniele Montenegro da Silva Barros, Rodrigo Dantas
1398 da Silva, Jailton Carlos de Paiva, Karilany Dantas Coutinho, Philippi
1399 Sedir Grilo de Morais, Juciano de Sousa Lacerda & Fernando Rocha de
1400 André, (2021), The relevance a technology ecosystem in the Brazilian
1401 National Health Service's Covid-19 response: the case of Rio Grande
1402 do Norte, Brazil, vol. 26, number 6, pp. 2035-2052, PMID: 34231717,
1403 DOI:10.1590/1413-81232021266.44122020
- 1404 [78] Dyshelly Nurkartika Pascapurnama, Aya Murakami, Haorile Chagan-
1405 Yasutan, Toshio Hattori, Hiroyuki Sasaki, & Shinichi Egawa, (2017),
1406 *Integrated Health Education in Disaster Risk Reduction: Lesson*
1407 *Learned from Disease Outbreak Following Natural Disasters in Indone-*
1408 *sia*, International Journal of Disaster Risk Reduction, N0 29, DOI:
1409 10.1016/j.ijdrr.2017.07.013
- 1410 [79] Alexandre R. Caitano, Cristine M. G. Gusmão, Sara Dias-Trindade, Ingridy
1411 M. P. Barbalho, Philippi Sedir G. Morais, Gleyson J. P. Caldeira-Silva,
1412 Manoel H. Romão, Janaína L. R. S. Valentim, Aline P. Dias, Joaquim L.
1413 M. Alcoforado, Carlos A. P. Oliveira, Karilany D. Coutinho, Maria C. F.
1414 D. Rêgo and Ricardo A. M. Valentim, (2022), Massive health education

- 1415 through technological mediation: Analyses and impacts on the syphilis
1416 epidemic in Brazil, *Frontiers in Public Health*, Public Health Education
1417 and Promotion, Volume 10 - 2022, DOI 10.3389/fpubh.2022.944213
- 1418 [80] Bouzidi Z., Boudries A., Amad M. (2021), *Deep Learning and Social Media*
1419 *for Managing Disaster: Survey*. In: Arai K. (eds) *Intelligent Systems and*
1420 *Applications*. IntelliSys 2021. Lecture Notes in Networks and Systems, vol
1421 294. pp. 12-30. Springer, Cham. [https : //doi.org/10.1007/978 – 3 – 030 –](https://doi.org/10.1007/978-3-030-82193-7_2)
1422 [82193 – 7.2](https://doi.org/10.1007/978-3-030-82193-7_2)
- 1423 [81] Bouzidi Z., Amad M. & Boudries A., (2020b), *Deep Learning-based Au-*
1424 *tomated Learning Environment using Smart Data to improve Corporate*
1425 *Marketing, Business Strategies, Fraud Detection in Financial Service and*
1426 *Financial Time Series Forecasting*, International Conference Managing
1427 Business through Web Analytics - (ICMBWA2020), Khemis Miliana Uni-
1428 versity, Algeria
- 1429 [82] Zair Bouzidi, Abdelmalek Boudries, Mourad Amad, (2022), Context-based
1430 Detecting Fake News, 31 October 2022, PREPRINT (Version 1) available
1431 at Research Square [<https://doi.org/10.21203/rs.3.rs-2194376/v1>]
- 1432 [83] Bouzidi Z., (2024), *Modeling economic environment monitoring to enhance*
1433 *economic tasks using deep learning with economical education on various*
1434 *sources*, International Conference on Nonlinear Mathematical Analysis and
1435 *Its Application*, Mai 14th – 15th, 2024 Bordj Bou Arréridj - Algeria
- 1436 [84] Kobayashi K., Narita A., Hirano M., Tanaka K., Katada T. & Kuwasawa
1437 K., (2008), *DIGTable: a tabletop simulation system for disaster educa-*
1438 *tion*, Proceedings of 6th International Conference on Pervasive Computing
1439 (Pervasive2008), p. 57-60
- 1440 [85] Lin C. K., Nifa F. A/ A., Musa S., Shahron Sy. A. & Anuar N. A.,
1441 (2018), *Challenges and opportunities of disaster education program among*

- 1442 *UUM student*, Proceedings of the 3rd International Conference on Ap-
 1443 plied Science and Technology (ICAST 18), Georgetown, Penang, Malaysia,
 1444 doi:10.1063/1.5055440
- 1445 [86] He R., Liu Y., Yu G., Tang J., Hu Q. & Dang J., (2016). *Twitter summa-*
 1446 *rization with social-temporal context*, World Wide Web, 20(2), pp. 267-290,
 1447 DOI:10.1007/s11280-016-0386-0
- 1448 [87] Lamsal R. & Kumar T. V. V., (2020), *Classifying Emergency Tweets for*
 1449 *Disaster Response*, International Journal of Disaster Response & Emer-
 1450 gency Management, 3(1), pp. 14-29, DOI:10.4018/IJDREM.2020010102
- 1451 [88] Rudra K., Sharma A., Ganguly N. & Ghosh S., (2018), *Characterizing and*
 1452 *Countering Communal Micro-blogs during Disaster Events*, IEEE Trans-
 1453 actions on Computational Social Systems, vol. 5, No. 2, pp. 403-417
- 1454 [89] Bouzidi Z., Boudries A. & Amad M., (2021b), *LSTM-based automated*
 1455 *learning with smart data to improve marketing fraud detection and finan-*
 1456 *cial forecasting*, 5th International Scientific Conference on Economics and
 1457 Management, (EMAN 2021), Serbia
- 1458 [90] Bouzidi Z., Boudries A. & Amad M., (2021), *Deep LSTM-based Automated*
 1459 *Learning Environment using Smart Data to Improve Awareness and Edu-*
 1460 *cation in Time Series Forecasting*, MC Medical Sciences journal, Volume
 1461 1, Issue 5, <https://doi.org/10.55162/MCMS.2021.01.034>
- 1462 [91] Bouzidi Z., Amad M. & Boudries A., (2022), *A. Deep Learning-based*
 1463 *Automated Learning Environment using Smart Data to improve Corpo-*
 1464 *rate Marketing, Business Strategies, Fraud Detection in Financial Ser-*
 1465 *vice and Financial Time Series Forecasting*, Springer Nature Springer
 1466 book_524532_1_En_ Sedkaoui
- 1467 [92] Torani S., Majd P. M., Maroufi S. S., Dowlati M. & Sheikhi R. A.,
 1468 (2019), *The importance of education on disasters and emergency: A re-*

- 1469 *view article*, Journal of Education and Health Promotion. vol. 8, No. 85,
1470 doi:10.4103/jehp.jehp 26218
- 1471 [93] Abdalla R. & Esmail M., (2019), *WebGIS for Disaster Management and*
1472 *Emergency Response*, Advances in Science, Technology & Innovation.
1473 Springer Nature Switzerland. ISBN 978-3-030-03828-1
- 1474 [94] Zaini N.A., Noor S.F.M. & Zailani S.Z.M., (2020), *Design and Develop-*
1475 *ment of Flood Disaster Game-based Learning based in Learning Domain*,
1476 International Journal of Engineering and Advanced Technology (IJEAT),
1477 9(4), pp. 679-685, DOI:10.35940/ijeat.C6216.049420
- 1478 [95] Tsai M.H., Chang Y.L., Kao C. & Kang S.C., (2015), *The effectiveness*
1479 *of flood protection computer game for disaster education*, Visualization in
1480 Engineering, vol. 3, No 1, 2015, p. 1-13
- 1481 [96] Smith S. & Ericson E., (2009), *Using immersive game-based virtual reality*
1482 *to teach fire-safety skills to children*, Virtual reality, Volume 13, p. 87-99
- 1483 [97] Mitsuhara H., Iwaka K., Kozuki Y., Shishibori M., Inoue T., Yamaguchi
1484 K., Takechi Y. & Morimoto M., (2016), *Penumbra Tourism: Place-based*
1485 *Disaster Education via Real-world Disaster Simulation*, 3rd International
1486 Conference on Information and Communication (ICT-DM), Vienna, Aus-
1487 tria
- 1488 [98] Fischer J., Jiang W. & Moran S., *AtomicOrchid: A mixed reality game*
1489 *to investigate coordination in disaster response*, Proceedings of the 11th
1490 International Conference ICEC-2012, p. 572-57
- 1491 [99] Bouzidi Z., (2022), Améliorations dans le Commerce et les Affaires, la
1492 Prévision des Séries Chronologiques Financières, la Détection des Fraudes
1493 et le Marketing grâce à l'Apprentissage en Profondeur, au Big Data et au
1494 Supercalcul, tout en Méfiant des Fausses Nouvelles : Défi, , 2^{emes} Assises
1495 de la Francophonie scientifique, organisées par l'Agence Universitaire de la
1496 Francophonie les 27 et 28 octobre 2022 au Caire (Égypte):

- 1497 [100] Bouzidi Zair, Boudries Abdelmalek, Amad Mourad, (2022), Smart Ed-
1498 ucation with Deep Learning and Social Media for Disaster Management
1499 in the pursuit of Environmental Sustainability While avoiding Fake News,
1500 12 September 2022, PREPRINT (Version 1) available at Research Square
1501 [<https://doi.org/10.21203/rs.3.rs-2042061/v1>]
- 1502 [101] Muttarak R. & Lutz W., (2014), *Is education a key to reducing vulnera-*
1503 *bility to natural disasters and hence unavoidable climate change?*, Ecology
1504 and Society 19(1): 42. <http://dx.doi.org/10.5751/ES-06476-190142>
- 1505 [102] Hui, F., (2023), Transforming educational approaches by integrating eth-
1506 nic music and ecosystems through RNN-based extraction. Soft Comput.,
1507 <https://doi.org/10.1007/s00500-023-09329-9>
- 1508 [103] Bouzidi Z., (2022), *Social Networks for Disaster Management*, Lambert
1509 Academic Publishing, ISBN:978-620-6-16121-9
- 1510 [104] Hamilton, Dale; Pacheco, Ryan; Myers, Barry; Peltzer, Brendan. 2020.
1511 kNN vs. SVM: A comparison of algorithms. In: Hood, Sharon M.; Drury,
1512 Stacy; Steelman, Toddi; Steffens, Ron, [eds.]. Proceedings of the Fire
1513 Continuum-Preparing for the future of wildland fire; 2018 May 21-24; Mis-
1514 soula, MT. Proceedings RMRS-P-78. Fort Collins, CO: U.S. Department of
1515 Agriculture, Forest Service, Rocky Mountain Research Station. p. 95-109
- 1516 [105] Behera RK, Das S, Rath SK, Misra S & Damasevicius R (2020) Compar-
1517 ative Study of Real Time Machine Learning Models for Stock Prediction
1518 through Streaming Data. JUCS - Journal of Universal Computer Science
1519 26(9): 1128-1147. <https://doi.org/10.3897/jucs.2020.059>